Nueva CHEM101
Fall 2026

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Quiz #2 Practice

Study Guide & Problem Set

Instructions: As with Quiz #2, use only a periodic table and a scientific calculator. Show your work, include units with every value, and draw Bohr models whenever a question asks about why an atom behaves the way it does.

Quiz 2 will assess the following:

- **[CK3]** Electrostatic forces: Explain how electrostatic interactions hold substances together, influence potential energy, and affect electron movement.
- **[CK4]** Periodic Trends: Describe the major periodic trends (e.g., electronegativity, ionization energy) and how they relate to the structure of the periodic table.
- **[S&P5]** CER: Construct a logically connected chain of evidence and reasoning that supports a claim.

Vocabulary you should know/be able to recognize if used in a question:

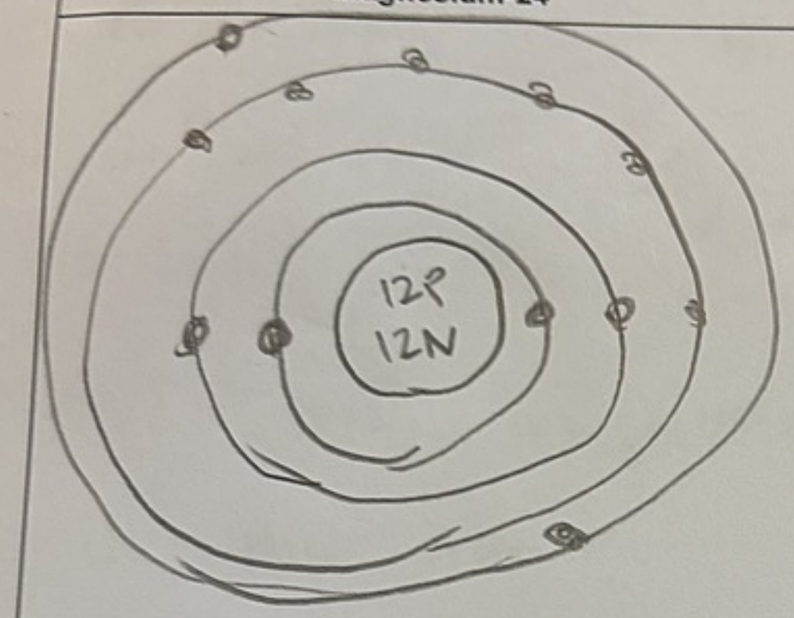
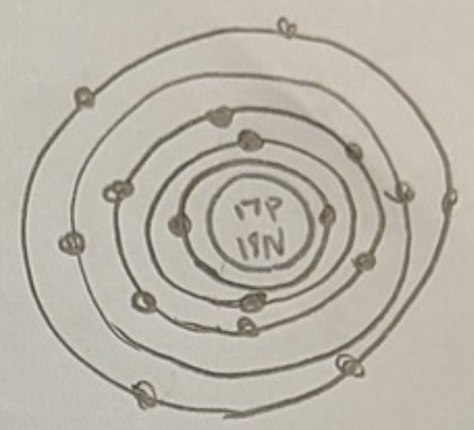
- Bohr model
- Electron shell (energy level)
- Valence electrons vs. inner (core) electrons
- Period vs. group
- Electrostatic attraction and repulsion
- Coulomb's law
- Electron shielding
- Effective nuclear charge (Z_{eff})
- Atomic radius
- Ionization energy (1st, 2nd, and successive)
- Electronegativity
- Periodic trend
- Ion (cation vs. anion)
- Claim, evidence, and reasoning (CER)

You should be able to:

1. Draw Bohr models of atoms and ions of the first 20 elements, labeling the protons, neutrons, and electrons in each shell;
2. Determine the number of valence electrons in an atom from its group number or Bohr model;
3. Calculate the core charge (protons − inner electrons) felt by an atom's valence electrons;
4. Use Coulomb's law to explain how charge and distance affect the attraction between a nucleus and its electrons, or between oppositely charged ions;
5. Explain how attraction relates to potential energy and to the energy needed to remove an electron;
6. Explain why electrons and ions release energy when they move to where they are attracted more strongly (e.g., in a battery);
7. Describe how atomic radius, ionization energy, and electronegativity change across a period and down a group;
8. Explain these trends using electron shells, shielding, effective nuclear charge (Z_{eff}), and Coulomb's law;
9. Interpret data tables and graphs of periodic trends, citing specific values with units; and
10. Write a CER that connects evidence and reasoning to a claim that answers a guiding question.

Practice Problems:

1. **[CK3 · Step 3]** Draw Bohr models of a magnesium-24 atom and a chlorine-35 atom. Label the number of protons and neutrons in a central circle that represents the nucleus and include the number of electrons in each shell. Then calculate the effective nuclear charge (Z_{eff}) felt by each atom's valence electrons. (n = 1 can hold up to 2 electrons, n = 2 & 3 can hold up to 8 electrons)

Magnesium-24	Chlorine-35
	

2. **[CK3 · Step 3]** Nitrogen and phosphorus are both in group 15.
   a. Which atom holds its valence electrons more tightly? Explain using effective nuclear charge and Coulomb's law.

Nitrogen holds holds its electrons for 2 key reasons. Coulombs law al effective nuclear charge. First coulombs law takes place because phosphrus valene electrons is on later shell making attraction better. In addition we see that both electrons have a +3 charge thrugh nuclear charge. Nitrogen closer so it holds tighter.

   b. Their 1st ionization energies are 14.53 eV and 10.49 eV. Which value belongs to which element? Justify your answer.

lower belongs to phosphrus because it dosnt hold the electron as tight it doesnt need as much energy

3. **[CK4 · Step 3]** Rank sodium (Na), magnesium (Mg), and potassium (K) from lowest to highest by:
   a. atomic radius, K, Na, Mg
   b. 1st ionization energy, and Mg, Na, K
   c. electronegativity. Mg, Na, K

[Step 4]: Justify each ranking using each element's position on the periodic table.

a) K is farthest because it's on highest shell, Mg more protons so pulls electrons

b) Na and Mg electrons same shell but Mg has more charge so harder to remove because more nuclear charge.

c) periodic trends in prev ex show attraction greater in this order

4. **[CK4 · Step 3]** The table shows 1st ionization energies across period 2.

Element	Li	Be	B	C	N	O	F	Ne
1st ionization energy (eV)	5.39	9.32	8.30	11.26	14.53	13.62	17.42	21.56

a) Using the periodic table to guide your reasoning, describe the overall trend using the format "As ___ increases, ___ ___."

As atomic # increase the ionization will increase because they are on same level but

b) Explain the overall trend using effective nuclear charge and Coulomb's law.

distance the same but charge diff T

bc they are on same level nuclear charge gets bigger as electrons fall down

5. **[CK3 · Step 3]** A sodium atom has electrons in shells 1, 2, and 3.
   a) Rank the potential energy of an electron in each shell from lowest to highest. Explain using Coulomb's law.

b) Explain why removing an electron from any atom always requires energy.

You have to countract columbs law of attrach

So you need a greate force to remove it

6. **[CK3 · Step 3]** The ions Na^+ (1.02 Å), Mg^{2+} (0.72 Å), and Al^{3+} (0.54 Å) all have exactly 10 electrons. Explain why the ions get smaller from Na^+ to Mg^{2+} to Al^{3+}, even though they all have the same number of electrons.

each atom has more protons .. when an atom has stron attraction it pulls in its electrons a lil tighter

So even tho same electrons they are pulled

closer

7. **[S&P5, CK4 · Step 3]** A student claims: "Chlorine has more protons than fluorine, so it should pull harder on shared electrons and have a higher electronegativity." Fluorine's electronegativity is 3.98, and chlorine's is 3.16. Evaluate this claim. Is it true, partially true, or false? If needed, correct the claim to be accurate.

false, fluorine may have less protons but its distance to the shared electrons less giving it more pull

8. **[CK3, CK4, S&P5 · Step 4]**

Scenario: Many satellites move using ion thrusters, which remove one electron from each atom of a gas and then use electric fields to fire the ions out the back. For years, most thrusters used xenon, but xenon is rare and expensive, so some companies have switched to argon. Every atom has to be ionized, so the energy needed to remove an electron is a major cost of running the thruster.

Property	Argon (Ar)	Krypton (Kr)	Xenon (Xe)
Atomic number	18	36	54
Atomic radius (Å)	0.71	0.88	1.08
1st ionization energy (eV)	15.76	14.00	12.13

Guiding question: Xenon has three times as many protons as argon. Why does it take *less* energy to remove an electron from xenon?

Write a CER that includes:

- **Claim:** one concise sentence that answers the question and names the cause.
- **Evidence:** at least three values with units from the table.
- **Reasoning:**
  - Draw Bohr models of argon and krypton, and calculate the core charge felt by each atom's valence electrons.
  - Explain how the data fit the trend down group 18.
  - Use shielding, Coulomb's law, and potential energy to explain why the extra protons don't make xenon's electron harder to remove.

[I recommend drafting and writing your CER in your science notebook!]

C: we tested 3 atoms to see their first ionization energy. Xenon has least ionization energy because the atomic radius

E: the evidence is a liner shape. Argon has atomic rad of 0.71 Krypton 0.88, Xenon 1.08, their ionization energy goes the opposite the. Argon, 15.76, krypton 14, Xenon 12.13

R: this principle can be explained through coulombs law the force is related on distance. Many the textbook in the table shows this fact

Optional extension problems

Use these practice problems to prepare for reassessment on the states of matter learning standard.

Step 3 Problems:

9. *Bromine melts at −7 °C and boils at 59 °C. Determine bromine's state at −20 °C, 25 °C, and 80 °C. Draw a particle diagram for each temperature, and write one sentence per diagram describing particle spacing, motion, and attractions.*

10. *You seal the tip of a syringe filled with air and push the plunger, then repeat with a syringe filled with water. Predict what happens in each case, and explain the difference using particle spacing.*